

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I S. Susannal Devapriyam (192525398) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S. Susannal Devapriyam

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

CSA1507 - Cloud Computing & Big Data Analysis.

CO1 - Assessment Tool - 1.

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Cloud Computing :

cloud computing is an on-demand service provider. which has storages, services, networking that can be accessed by users on their basis.

Characteristics:

- Broad Network Access: User can access the service from clouds through the internet without having to manage infrastructure.
- On - Demand self service: Users can use the software without the cloud service provider's permission.
- Rapid scalability: Resources can be increased or decreased quickly based on user's preference.
- Resource pooling: cloud computing helps users on the world to access the resources simultaneously by also protecting or securing user's data.

How it differs from Traditional on-premise computing:

- Traditional on-premise computing needs service provider's acceptance to provide services, networks and resources.
- cloud computing provides data, resources and software all through the internet without acceptance of service provider.

Cloud computing has evolved from basics in all the ways listed as follows:

- Hardware evolution:

The computers have evolved from large mainframes into personal computers, Virtualization softwares with high data processing and Reliable resources.

- Internet Software evolution:

The computer resources is being used throughout the wide are network with improvement in LAN (Local Area Network) and Wan (Wide area Networks) to provide service to the users.

- Distributed Systems:

By the use of grid computing and cluster computing the certain task is getting completed by several computers working together. results in faster problem solving through resources.

- Service Delivery:

By providing software as a service the modes such as IaaS (Infrastructure as a service), PaaS (platform as a service), SaaS (Software as a service), XaaS (everything as a service) has be developed.

- Such Evolutions made cloud computing way more reliable and scalable in providing services to the users.

3. Server Virtualization:

Server Virtualization creates many virtualization machines on a single server to share resources in cloud computing services.

Role in cloud computing:

- Improves hardware utilization.
- Supports resource sharing.
- High Reliability and Scalability.
- provides rapid scaling.
- Helps in Backup and recovery.

Differences:

Server Virtualization	Cloud Computing
• Technology • Focuses on Hardware utilization • Share resources through physical server	• Service provider model. • Focuses on service providing and delivery • Uses virtualizer to provide & share service, resources.

- It uses a server that combines all the virtualizer to provide hardware utilization and reliable scaling called the 'Hypervisor'.

Data Centers:

Data centers store large amounts of data in the storages.

It also has cooling systems, software resources, resource accessing equipments and also helps in data recovery.

Role of Datacenters:

- Host web applications and web services.
 - provides Data storages
 - uses firewalls and backups for recovery.
 - High data storing capacity.
 - Has high Scalability to enhance the collection of data through services.
 - It has the facility to store data in large amounts to access it later.
- Hence the Data Centers plays an important role as a Back Bone to the cloud computing that stores the data and ~~stores~~ provides Infrastructure management as a service to the users to access the resources, software and web applications over the internet.
- It is also used to recover the data and failed systems using firewalls and backup.

Service Model	Description	Example.
• IaaS. Infra structure as a Services	provides resources, networking and computers, user have to manage the providance of applications or softwares.	Amazon EC ₂
• PaaS. platform as a Service.	User is provided with the platform, & also provides software. user manages development and deployment of Softwares without have to manage Infra structure.	Google App Engine
• SaaS Software as a Services	It provides ready to use software over the Internet. User is provided with the actual application and software to work on it and save tools.	Google Docs, GMail.
• XaaS. Everything as a Service	provides everything to the user, they have to use to learn it includes AI and web softwares and documents	Microsoft Azure.